

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1755

Sheridan Alte 138 kV
December 17, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1755. The Interconnection Customer proposes a 74.9 MW solar plus storage facility interconnecting at the Sheridan Alte 138 kV substation in the Virginia area. The request was received on May 21, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 493 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$452,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
De Kalb – McClain Energy Facility 138kV	Colton – Coal 138kV L	116.8	220	8.2
De Kalb – Oxnard 138kV	Colton – Coal 138kV line	112.4	223	8.2
Hinton Ap – Oxnard 138kV	Colton – Coal 138kV	106.0	229	8.2
Huntington Court – Bonsack 138kV	Craven County Wood Energy Lp – Lone Tree 138kV	101.8	352	25.1
Vc Porterdale Hydroelectric – Bonsack 138kV	Craven County Wood Energy Lp – Lone Tree 138kV	101.7	386	25.1
Blue Ridge – Neely 138kV line	Craven County Wood Energy Lp – Lone Tree 138kV	101.3	358	36.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
De Kalb – McClain Energy Facility 138kV	Rebuild line with higher-capacity conductor	\$4,238,000	2% (\$70,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
De Kalb – Oxnard 138kV line	Rebuild line with higher-capacity conductor	\$11,125,000	2% (\$184,000)
HintonAp – Oxnard 138kV	Reconductor line	\$1,698,000	2% (\$28,000)
Huntington Court – Bonsack 138kV	Reconductor line	\$601,000	5% (\$30,000)
Vc Porterdale Hydroelectric – Bonsack 138kV Line	Reconductor line	\$1,080,000	5% (\$53,000)
Blue Ridge – Neely 138kV line	Reconductor line	\$1,566,000	6% (\$87,000)
	Total allocated to project		\$452,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.